Zaid Liyakatali Sarguroh

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Date of birth: 15/04/2000 **Nationality:** Indian

ABOUT ME

A Model-Based Development Engineer with over 2 years of experience in embedded software development for automotive control systems. Passionate about embedded control technologies, particularly in autonomous driving and electrification, I'm eager to deepen my expertise through graduate studies and contribute to the future of the industry.

EDUCATION AND TRAINING

[2017 – 2021] **Bachelor of Technology in Mechanical Engineering**

National Institute of Technology Rourkela, India

Final grade: 7.78/10

- **Relevant Courses:** Mechatronics, System Dynamics and Control, Electrical Machines, Mechanical Vibration

[2016 – 2017] **XII Grade**

International Indian School Riyadh, Saudi Arabia

Final grade: 87.2%

[2014 – 2015] **X Grade**

International Indian School Riyadh, Saudi Arabia

Final grade: 10/10

WORK EXPERIENCE

Devise Electronics Pvt. Ltd.

City: Pune | **Country:** India

[01/04/2025 – 01/08/2025] **Senior Engineer**

- ASW development for EVCC involving charging communication, session control and fault handling.
- BSW development support to handle vehicle diagnostics requests over UDS.

[01/09/2023 – 31/03/2025] **Junior Engineer**

- Developed supervisory controls software for electric commercial vehicles using MATLAB/Simulink.
- Implemented torque arbitration strategy between VCU and ZF EBS to improve braking performance.
- Integrated ECUs with CAN and J1939, including implementation of transport protocol, address claiming and DMx messages.
- Supported low-level code development in C for hardware integration.

[01/03/2023 – 31/08/2023] **Intern**

- Assisted in developing electric powertrain control logic, including one-pedal driving and torque generation using MATLAB/Simulink.
- Created and validated test cases (MIL/HIL) for control algorithm performance and reliability.
- Conducted CAN data acquisition and analysis with BUSMASTER, Vector tools and Python scripting.

SKILLS

MATLAB | Simulink | Stateflow | C | C++ | Python | Vector CANdb++ | Vector CANalyzer | BUSMASTER | Controller Area Network (CAN) | SAE J1939 | Git

PROJECTS

Design and Implementation of a Closed-Loop Electronic Throttle Control System

- Developed plant model of a Bosch electronic throttle body using Simulink and Simscape.
- Developed and integrated a PID controller for precise throttle positioning and improved response.
- Validated the control software on a GCM196 ECU using New Eagle Raptor Controls Platform.

Active Suspension System Modeling and Control for a Half-Car Model

- Developed a mathematical model of a half-car active suspension system in MATLAB/Simulink.
- Analyzed suspension response to parameter variations impacting ride quality and handling.
- Implemented and tuned a PID control strategy to actively regulate suspension behaviour.

HONOURS AND AWARDS

Scholarship Programme for Diaspora Children (SPDC) Awarding institution: Ministry of External Affairs, Government of India

- Awarded a merit-based scholarship throughout the bachelor's programme.

CERTIFICATIONS

[14/02/2022 – 13/05/2022] **Hands-On Controls (IEEE Certified, Online) - DORLECO**

- Gained experience in modern techniques used for modelling automotive system controls.

[27/06/2020 – 25/07/2020] **Vocational Training (Online) - National Thermal Power Corporation (NTPC) Sipat**

- Conducted research on the thermodynamics and control operation of AUSC thermal power plants.

LANGUAGE SKILLS

English, German (A2)